

Foreground Masking Canonical Tools Reference

Updated naming convention and launch guidance for the foreground masking interactive and batch tools.

Naming Convention

There are now four canonical interactive entry points and four canonical batch entry points. Each filename starts with interactive or batch, then names the foreground-detection method, then names the masking technique.

Pattern: interactive_<method>_<technique>.py or batch_<method>_<technique>.py

Methods are spike_gate and toy_objects. Techniques are SEP and MTOjects.

Interactive Tools

Canonical file	Method	Technique	Default behaviour
interactive_spike_gate_SEP.py	Spike Gate	SEP	Opens SEP + Spike Gate GUI and loads latest sep_spike_optimisation_best.json by default.
interactive_toy_objects_SEP.py	Toy Objects	SEP	Opens SEP toy-object recovery GUI and loads latest sep_toy_object_optimisation_best.json by default.
interactive_spike_gate_MTObj ects.py	Spike Gate	MTOjects	Opens MTOjects + Spike Gate GUI and loads latest mtojects_spike_optimisation_best.json by default.
interactive_toy_objects_MTObj ects.py	Toy Objects	MTOjects	Opens MTOjects toy-object recovery GUI and loads latest mtojects_parameter_optimisation_best.json by default.

Batch Tools

Canonical file	Method	Technique	Default behaviour
batch_spike_gate_SEP.py	Spike Gate	SEP	Runs all-galaxy SEP batch using latest SEP Spike Gate optimisation unless --best-json is supplied.
batch_toy_objects_SEP.py	Toy Objects	SEP	Runs all-galaxy SEP batch using latest SEP toy-object optimisation unless --best-json is supplied.
batch_spike_gate_MTObj ects.py	Spike Gate	MTOjects	Runs all-galaxy MTOjects batch using latest MTOjects Spike Gate optimisation unless --best-json is supplied.
batch_toy_objects_MTObj ects.py	Toy Objects	MTOjects	Runs all-galaxy MTOjects batch using latest MTOjects toy-object optimisation unless --best-json is supplied.

Default Parameter Loading

Interactive tools load the most recent matching optimisation result by default. For example, `interactive_spike_gate_SEP.py` searches the SEP Spike Gate optimisation folder only; it will not accidentally load a newer toy-object result.

All tools still accept `--best-json`, which overrides the automatic latest-result lookup. Use `--pc Laptop` or `--pc Desktop` to select the matching Dropbox/Erwin path tree.

The Spike Gate interactive defaults are: Spike Gate detects on residual; SEP or MTOBjects detects on original.

Example Commands

```
python "Foreground Masking\Interactive tools\interactive_spike_gate_SEP.py" --pc Desktop
```

```
python "Foreground Masking\Interactive tools\interactive_toy_objects_SEP.py" --pc Desktop
```

```
python "Foreground Masking\Interactive tools\interactive_spike_gate_MTOBjects.py" --pc Desktop --mtobjects-root "C:\Users\gordo\Documents\Github\mtobjects"
```

```
python "Foreground Masking\Interactive tools\interactive_toy_objects_MTOBjects.py" --pc Desktop --mtobjects-root "C:\Users\gordo\Documents\Github\mtobjects"
```

```
python "Foreground Masking\Batch tools\batch_spike_gate_SEP.py" --pc Desktop --require-best-json
```

```
python "Foreground Masking\Batch tools\batch_toy_objects_SEP.py" --pc Desktop --require-best-json
```

```
python "Foreground Masking\Batch tools\batch_spike_gate_MTOBjects.py" --pc Desktop --mtobjects-root "C:\Users\gordo\Documents\Github\mtobjects"
```

```
python "Foreground Masking\Batch tools\batch_toy_objects_MTOBjects.py" --pc Desktop --mtobjects-root "C:\Users\gordo\Documents\Github\mtobjects"
```

Legacy Names

Older scripts are retained as compatibility entry points for existing scheduled jobs and notebooks. New work should use the canonical names below.

Older entry point	Use this canonical tool
<code>interactive_sep_spike_gate_parameter_tester.py</code>	<code>interactive_spike_gate_SEP.py</code>
<code>interactive_sep_toy_object_tester.py</code>	<code>interactive_toy_objects_SEP.py</code>
<code>interactive_mtobjects_spike_gate_parameter_tester.py</code>	<code>interactive_spike_gate_MTOBjects.py</code>
<code>interactive_mtobjects_toy_object_tester.py</code>	<code>interactive_toy_objects_MTOBjects.py</code>
<code>batch_sep_all_galaxies.py --source spike-gate</code>	<code>batch_spike_gate_SEP.py</code>
<code>batch_sep_all_galaxies.py --source toy-object</code>	<code>batch_toy_objects_SEP.py</code>
<code>apply_optimised_mtobjects_all_galaxies.py --source spike-gate</code>	<code>batch_spike_gate_MTOBjects.py</code>
<code>apply_optimised_mtobjects_all_galaxies.py --source toy-object</code>	<code>batch_toy_objects_MTOBjects.py</code>

Implementation Notes

- `canonical_tool_helpers.py` contains the latest-best-json lookup shared by the canonical wrappers.
- SEP batch execution remains implemented by `batch_sep_all_galaxies.py`.
- MTOBjects batch execution remains implemented by `apply_optimised_mtobjects_all_galaxies.py`.

- SEP Spike Gate GUI execution remains implemented by `interactive_sep_spike_gate_parameter_tester.py`.
- MTOBJECTS Spike Gate GUI execution remains implemented by `mtobjects_spike_gate_processing.py`.
- Toy-object GUI execution remains implemented by `toy_object_interactive_core.py`.

Systematic documentation review - 16 August 2026

Review classification: Updated - canonical launch paths and batch-output conventions

Updates made: Example commands were corrected to include the Interactive tools and Batch tools folders. The automatic `\_clean` suffix for the validated 40 calibration galaxies and the aligned eight-panel Toy Objects batch layout were documented.

Current interpretation: Future SEP and MTOBJECTS Toy Objects batches read `CleanGalaxies.txt`, require 40 unique names, and add `\_clean` before `.png` for those calibration galaxies.

Authoritative combined outcome: SEP and MTOBJECTS Toy Objects Optimisation Summary and Conclusions.docx. Full-precision results: Foreground Masking Optimisation Results.xlsx.

- Aligned Toy Objects batch layout: original / original plus toys; mask / recovered image with green correct and red incorrect outlines; original / processed isophotes; original / processed bar-major profile.
- Calibration filenames: reports for the 40 validated `CleanGalaxies.txt` entries end in `\_clean.png`; the remaining reports keep their ordinary suffix.